IPR MANAGEMENT

Intellectual Property Rights Management

End Term Notes

Final Year LLB

Part One: Measuring and Extracting Value

Part Two: Portfolio Management

Part Three: Innovation and Intellectual Property Protection

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Part One: Measuring and Extracting Value

1.1 Meaning and Importance of Measuring Intangible Assets

Meaning

Intangible assets are resources that have no physical form but are expected to bring future benefits to a business. For a law student, the core of intangible assets is the set of **legal rights** recognised by Indian law: patents under the Patents Act, 1970, copyright under the Copyright Act, 1957, trade marks under the Trade Marks Act, 1999, designs under the Designs Act, 2000, geographical indications under the Geographical Indications of Goods (Registration and Protection) Act, 1999, and confidential information protected through contract and equity. Management takes a wider view and also counts know how, software, data, brand reputation, customer relationships and the skills of employees, even when no registered right exists.

Ind AS 38 gives the accounting meaning. An intangible asset is an **identifiable non monetary asset without physical substance**. It is identifiable if it can be separated from the business or if it arises from **contractual or legal rights**. The business must **control** it, which in practice usually means owning a legal right or a contractual right, and it must expect **future economic benefits** from it. The law of IP therefore decides much of what accounting can recognise: a right that is not legally secured is often not controlled, and what is not controlled cannot be an asset.

Features

Intangible assets behave differently from machinery or land. Knowledge can be used by many people at the same time without being used up, and once created it can be copied at very low cost. Others can easily imitate it, so its value depends on legal exclusivity created by IP law or on secrecy. Its future returns are uncertain, it is rarely bought and sold in open markets, and its value depends on complementary assets such as manufacturing, distribution and brand. It can become obsolete quickly. Much of it sits in the minds of employees, who are not owned by the firm and may leave. Granstrand adds that the value of an IP portfolio is not simply the total of the values of its separate rights, because rights support and overlap each other.

What Measuring Involves

Measuring intangible assets happens at four levels. The first is **identification**, which means listing what the firm has, usually through an IP register. The second is **classification**, which groups assets by type, technology, business or legal status. The third is **quantification**, which uses non

money indicators such as the number of patents, citations, staff turnover or customer retention. The fourth is **valuation**, which puts a money value on an asset for a stated purpose and date.

Importance

Measurement matters both legally and commercially. Management needs it to identify sources of competitive advantage, to divide budgets between R&D, marketing and IP, and to decide which rights to renew, license, sell or abandon. Legal transactions depend on it: licences, assignments, mergers, joint ventures and technology transfer agreements need a price, and courts need a basis for damages in infringement suits. Lenders need it before accepting IP as security, tax authorities need it for royalties and transfer pricing, and accountants need it for acquisitions and impairment tests. Investors need it to judge knowledge companies, and better information can lower the cost of capital. Measurement also reveals risks, such as dependence on one patent, one licence or a few key inventors. In short, what is not measured is usually not managed.

Why Measuring Is Difficult

Measurement is hard because much knowledge is tacit and undocumented, comparable market transactions are rare, benefits lie in an uncertain future, and the contribution of one asset is difficult to separate from others. Results depend on assumptions about royalty rates, discount rates and useful life. There is also **legal uncertainty**: a patent may be revoked, a mark may become generic and a trade secret may leak.

Accounting Recognition and Management Measurement

Accounting recognition and management measurement are different exercises. Accounting aims at reliable financial statements and follows Ind AS 38 or AS 26. Under Ind AS 38, research cost is always an expense (para 54). Development cost becomes an asset only when six conditions are met: technical feasibility, intention to complete, ability to use or sell, probable future benefits, adequate resources and reliable measurement of cost (para 57). Internally generated brands, mastheads, publishing titles and customer lists are never recognised (para 63), and internally generated goodwill is never recognised (para 48). Bought intangibles are recorded at cost and amortised over their useful life if that life is finite. Companies outside Ind AS follow AS 26, which takes a similar approach. Management measurement is voluntary, covers all intangibles including human capital, and aims at better decisions rather than audited reliability.

Because of these rules, the balance sheet shows few internally created IP rights, so the **market value** of a knowledge firm is often far above its **book value**.

Market to book ratio = Market value of equity ÷ Book value of net assets

If market value is ₹10,000 crore and book value ₹2,000 crore, the ratio is 5. The gap partly reflects unrecorded intangibles, but it also reflects growth expectations and market mood, so it cannot be treated as an exact value of the firm's IP.

1.2 Public Policy on Measuring Intangible Assets and Its Implications

Meaning and Concern of the State

Public policy on this subject consists of the rules made by government, regulators and standard setters on how intangible assets are recognised, valued, disclosed, audited, taxed, financed and counted in national statistics. The State is concerned because intangibles now form a large share of business value, investors need reliable information, firms rich in IP but poor in physical assets struggle to borrow, R&D benefits society beyond the firm, IP can be shifted easily to low tax countries, and IP and data can create market power.

The Policy Choice

The central policy choice is between **relevance and reliability**. Recognising more intangibles would make accounts more realistic, help investors and reduce the cost of capital for knowledge firms. Against this, values of intangibles are often estimates that are hard to audit, can be manipulated to inflate balance sheets, and may rise and fall sharply. Most standard setters therefore keep recognition strict and encourage better **disclosure** of non money indicators and narrative explanation.

Indian Legal and Regulatory Framework

India follows a conservative recognition rule through **Ind AS 38**, notified under the Companies (Indian Accounting Standards) Rules, 2015, and **AS 26** for other entities. Valuations required under company law must be done by **registered valuers** under s. 247 of the Companies Act, 2013 and the Companies (Registered Valuers and Valuation) Rules, 2017, and the ICAI Valuation Standards 2018 include **VS 302 on intangible assets**. The Companies (Auditor's Report) Order, 2020 requires the auditor to report whether the company keeps proper records of its intangible assets and to report on any revaluation. Tax law allows **depreciation on acquired intangibles** such as know how, patents, copyrights, trade marks, licences and franchises, but not goodwill (Income Tax Act, 2025, s. 33(1)(b)), and taxes royalty from patents developed and registered in India at a concessional 10 per cent (s. 194(1)). Finance law recognises IP as property that can be secured: the **SARFAESI Act, 2002** includes intangible assets in the definition of property, and security interests over them are registered with **CERSAI**. The **National IPR Policy, 2016** lists commercialisation of IPR as an objective, including valuation and IP based financing, and the SIPP scheme supports start ups with IP filings.

Implications

Under this framework, investors must estimate intangible value themselves and may misprice firms. Lenders find little security on the balance sheet, so IP backed lending remains small. Start ups report losses because R&D is expensed and therefore depend on equity. Government must use tax incentives and grants to encourage R&D. Tax authorities and taxpayers dispute royalty

values in transfer pricing. Boards manage valuable assets that do not appear in the accounts, and national statistics may understate investment in knowledge.

International Position

The International Accounting Standards Board is reviewing IAS 38 as a research project, but no change had been made by 7 November 2026; any change would reach India only through Ind AS. Integrated reporting encourages firms to report on six capitals: financial, manufactured, intellectual, human, social and relationship, and natural. WIPO runs a programme on IP finance and publishes country reports. The overall trend is towards better disclosure rather than wider recognition.

1.3 Intellectual Capital Accounting as a Management Tool

Meaning and Components

Intellectual capital (IC) is the knowledge, skills, systems and relationships of a firm that can be converted into value. Granstrand divides a firm's capital into physical, financial and intellectual capital, and treats IC as including IP rights, human competence and goodwill. IC has three components. **Human capital** is the knowledge and skill of employees, which the firm does not own and which leaves when people leave. **Structural capital** is what remains in the firm, such as processes, databases, systems, culture and IP rights. **Relational capital** is the value of relationships with customers, suppliers and partners, including brand and reputation. The Skandia scheme expresses this as: market value equals financial capital plus intellectual capital; intellectual capital equals human capital plus structural capital; structural capital equals customer capital plus organisational capital; and organisational capital equals innovation capital plus process capital.

Meaning of IC Accounting

IC accounting is the identification, measurement and reporting of intellectual capital, mainly for internal management. It is voluntary and is not part of the statutory financial statements. From a legal point of view, it is useful because it connects IP rights with the people, systems and relationships that make them valuable, and it supports IP audits, due diligence and valuation. Where a company chooses to publish IC information in a prospectus, the information must not be untrue or misleading, since ss. 34 and 35 of the Companies Act, 2013 impose liability for misstatements in a prospectus.

Main Frameworks

The **Balanced Scorecard** of Kaplan and Norton measures performance from four perspectives (financial, customer, internal process, and learning and growth) and links them through cause and effect. The **Skandia Navigator** of Edvinsson and Malone uses five focus areas: financial, customer, process, renewal and development, and human, with human capital at the centre.

Sveiby's **Intangible Assets Monitor** measures external structure, internal structure and the competence of people, each through indicators of growth, efficiency and stability. Brooking's **Technology Broker** divides IC into market assets, human centred assets, IP assets and infrastructure assets. The **IC Index** of Roos combines indicators into one index and tracks changes over time. Sveiby also groups measurement methods into four families: direct methods that value each part of IC, market capitalisation methods such as market value minus book value, return on assets methods that capitalise excess profit, and scorecard methods that use non money indicators.

Key Formulas

Tobin's q = Market value of the firm ÷ Replacement cost of its assets

A q above 1 suggests that the market sees valuable intangibles that replacement cost does not capture.

VAIC = HCE + SCE + CEE, where HCE = VA ÷ HC, SCE = (VA minus HC) ÷ VA and CEE = VA ÷ CE

Here VA is value added, HC is employee cost and CE is capital employed. If VA is 200, HC is 80 and CE is 400, then HCE is 2.5, SCE is 0.6 and CEE is 0.5, so VAIC is **3.6**. Under the **Calculated Intangible Value** method, if a firm earns average pre tax profit of 90 on tangible assets of 500 while the industry earns 12 per cent, the excess profit is 90 minus 60, or 30. After 25 per cent tax this is 22.5, and dividing by a 15 per cent cost of capital gives a CIV of **150**.

Use and Limits

IC accounting helps managers form strategy, allocate resources, measure performance beyond profit, manage knowledge and risks, and communicate with investors. Its weaknesses are that there is no common standard, indicators can be manipulated, cause and effect is assumed rather than proved, and money values depend heavily on assumptions.

1.4 Extracting Profits from Intellectual Capital: Policy and Practice

Creation and Extraction of Value

Value creation is the production of new knowledge and inventions. **Value extraction** is the conversion of that knowledge into money or market power. Many firms create valuable knowledge but earn little from it because they lack a clear IP policy, legal protection or complementary assets.

Teece's Framework and the IP Value Hierarchy

Teece explains that the innovator profits only if three conditions favour it. The first is the **appropriability regime**: strong legal protection, such as valid patents or well kept secrets, allows the

innovator to profit, while weak protection helps imitators. The second is the **dominant design**: once a standard design emerges, competition moves to price and scale. The third is **complementary assets** such as manufacturing, distribution, service and brand; if imitators own these and protection is weak, they may capture most of the profit. Harrison and Sullivan describe an **IP value hierarchy** in which firms progress from using IP defensively, to controlling IP costs, to treating IP as a profit centre, to integrating IP with business strategy and finally to a visionary stage in which IP shapes the future of the firm and the industry.

Methods of Extraction and Their Legal Structure

The simplest method is **own use**, where IP keeps competitors out and protected features, designs and marks support higher prices. **Licensing** allows others to use IP for a royalty while the licensor keeps ownership; a patent licence must be in writing and registered to be relied upon (Patents Act, ss. 68 and 69), and a copyright licence must be in writing (Copyright Act, s. 30). **Assignment** transfers ownership for a price, and requires writing and registration for patents, writing for copyright (s. 19) and registration for trade marks (Trade Marks Act, s. 45). **Franchising** licenses a business system with marks and know how, and the franchisor must control quality to protect the mark. **Technology transfer** combines patent licences with know how and technical assistance.

Collaborative methods include **joint ventures**, where partners contribute IP to a new company, **strategic alliances** based on contracts, **research collaborations** with universities and firms, and **open innovation**, where firms buy in outside ideas and sell out unused ones. Financial methods include using IP as **security for loans**, selling future royalties and **securitising** royalty streams. **Enforcement** through infringement suits can produce damages or licences. **Defensive publication** keeps a field free by making an invention public so that nobody can patent it. Patents included in **technical standards** earn royalties from all implementers, **patent pools** let several owners license related patents together, and **copyright societies** license works collectively. Firms may also build **data and software based business models**, create **spin off companies**, **sell or donate non core patents** to save renewal fees, and use patents as **bargaining chips** to obtain cross licences and freedom to operate.

Choosing the Route

A firm that owns manufacturing and distribution usually earns most by using its IP itself. A firm lacking capital or market access does better through licensing or partnership. Strong legal protection makes licensing safer, while weak protection favours speed, secrecy and own use. Core assets are kept, and non core assets are licensed or sold. The choice must also account for tax, competition law and foreign exchange rules for cross border payments.

Policy and Legal Limits

Granstrand distinguishes a **policy**, which is a long lasting general guideline, from a **strategy**, which is a specific plan of action. His model IP policy makes business managers responsible for building and using IP, uses several types of protection for each business, seeks a strong patent position in core businesses, requires checking others' patents before entering a new field, encourages early filing ("speed to patent"), adopts open licensing for a proper royalty with active cross licensing, and requires respect for others' IP.

The law also limits how value can be extracted. Section 3(5)(i) of the Competition Act, 2002 protects only **reasonable conditions** needed to protect IP rights. Section 140 of the Patents Act makes certain restrictive conditions in patent licences void, such as tying the licensee to buy unpatented goods from the licensor, exclusive grant backs, preventing challenges to validity and coercive package licensing. A patent that is not worked in India may attract a **compulsory licence** under s. 84 after three years from grant, which is why the statement of working in Form 27 is important. Cross border royalty payments must also comply with foreign exchange and tax rules.

1.5 Intellectual Capital Measurement for Start Ups

Why IC Matters

In a start up almost all value is intangible: the founders, the technology, the business idea and early customers. Investors therefore judge the company mainly on its intellectual capital. Measuring it is difficult because there is no past performance and often no revenue, uncertainty is high, the business depends on a few founders, patents are usually pending rather than granted, and ownership of IP is often unclear. Granstrand warns that founders tend to overvalue their own ideas, which he calls "hubris", and notes the mistaken belief among small firms that one good patent is enough.

Legal Foundation

For a law student, the first question in measuring a start up's IC is **ownership**. A patent application may be filed by the true and first inventor or by an assignee (Patents Act, s. 6), so founders must assign their inventions to the company in writing, and the assignment should be registered (ss. 68 and 69). Copyright in work made by an employee in the course of employment generally belongs to the employer (Copyright Act, s. 17), but work done by outside consultants or freelancers belongs to them unless assigned in writing (s. 19). Confidentiality agreements protect know how, a trade mark search avoids conflicts, and a freedom to operate check shows whether the product may infringe others' patents.

What to Measure

A start up's IC should be measured through its team (founders' experience, completeness of the team, advisers), its structural capital (stage of the prototype, documented technology, data and processes), its IP position (ownership, filings, countries, claim scope, secrecy controls and freedom to operate), its relational capital (pilot customers, partners and letters of intent) and its market traction (market size, users, revenue and growth).

Valuation Methods

The **scorecard method** starts with the average value of similar funded start ups and adjusts it by weighted factors: team 30 per cent, size of opportunity 25 per cent, product and technology 15 per cent, competition 10 per cent, sales channels 10 per cent, need for further funding 5 per cent and others 5 per cent. If the average value is ₹20 crore and the weighted score is 1.155, the value is **₹23.1 crore**. The **Berkus method** gives value for five risk reducing elements: a sound idea, a prototype, a good team, strategic relationships and a product launch or sales. The **venture capital method** works backwards from the expected exit value:

Post money value = Expected exit value ÷ (1 + target return)^{years}; Pre money value = Post money value minus investment

If the exit value is ₹200 crore in five years, the investor targets 50 per cent a year and invests ₹10 crore, the post money value is 200 ÷ 7.594, or ₹26.34 crore, the pre money value is ₹16.34 crore and the investor's share is about **38 per cent**. The **First Chicago method** weights values in best, normal and worst cases by their probabilities.

Indian Support

Start ups recognised by DPIIT receive help with IP filings. The SIPP scheme provided facilitators and fee support and was extended up to 31 March 2026. Start ups pay lower patent fees and may request expedited examination of patent applications.

1.6 The IP Portfolio as a Competitive Tool

Meaning and Functions

An **IP portfolio** is the whole group of IP rights that a firm owns or controls, managed together. Legally, it is a bundle of territorial rights with different terms, owners and conditions. Because the rights support each other, a portfolio gives more power than single rights. It keeps competitors out of markets, differentiates products, strengthens the firm's position in licensing negotiations, protects freedom to operate because others hesitate to sue a firm that can sue back, blocks rivals' research directions, gives influence in technical standards, earns licensing income, provides access to others' technology through cross licences, attracts investors and lenders, and reveals competitors' strategies.

Why Firms Patent

Granstrand's survey of large Japanese corporations found that firms value patents for **protection** (of products above all, of processes, and to threaten counter attack), for **bargaining** (licensing, cross licensing, R&D cooperation and standard setting), for **image**, and least for **internal** reasons such as motivating inventors or measuring R&D. Large portfolios create **retaliatory power**: firms depend on each other's patents, so a strong portfolio deters attack. A party that holds one patent but has no manufacturing to be counter attacked can make demands on large firms; Granstrand calls such parties "patent extortionists". Offensive blocking secures **market freedom**, while defensive blocking secures **design freedom**.

Portfolio Strategies

Granstrand describes several patent strategies. **Ad hoc blocking** uses a few patents for one use and is easy to invent around. A **strategic patent** is one patent that is too costly for rivals to invent around. **Blanketing or flooding** covers a field with many patents. **Fencing** places a row of patents across several research paths. **Surrounding** places many small patents around a rival's key patent to force a cross licence, and a **patent network** combines different patents deliberately. Patenting may be **sporadic**, covering only key inventions, or **continuous**, filing throughout the product life, which costs more but gives wider and longer protection.

A firm facing a blocking patent has legal and commercial options. It may oppose the patent before grant (Patents Act, s. 25(1)) or within one year from publication of the grant (s. 25(2)), seek its revocation (s. 64), apply for a compulsory licence where the conditions of s. 84 are met, invent around it, buy or license it, offer a cross licence, wait for expiry or leave the field.

Size Is Not Value

A few patents usually carry most of the value of a portfolio. Its real strength depends on validity, claim scope, use in products, countries covered, remaining life, fit with strategy and cost of maintenance. Unused patents also carry legal risk, since non working can support a compulsory licence. Granstrand records that leading Japanese firms moved from quantity to quality by seeking "strategic patents".

1.7 The Role of the Intellectual Asset Manager

Meaning and Evolution

The **intellectual asset manager**, also called IP manager or chief IP officer, manages IP as a business asset by identifying, protecting, using and defending it in line with business strategy. Granstrand describes four stages in the evolution of corporate IP management. At the first stage IP is ignored. At the second, inventors are rewarded and IP is left to the legal department. At the third, patenting becomes selective, licensing happens when needed and patent positions are re-

viewed. At the fourth, IP becomes part of business strategy, engineers rotate through the IP department and licensing is based on business needs. The role therefore moves from **legal administration** to **business strategy**.

Responsibilities

The IA manager frames the firm's IP policy and runs an IP committee, aligns IP strategy with business strategy, collects invention disclosures and maintains an IP register, decides which form of protection to seek and in which countries, reviews the portfolio and decides on renewal, licensing or abandonment, and plans IP budgets. The manager negotiates licences and sales, drafts and manages IP clauses in employment, consultancy, research and supply contracts, checks freedom to operate, monitors infringers, coordinates valuation, due diligence and audit, keeps records and deadlines, trains staff, runs inventor reward schemes and links the legal, R&D, finance, tax and business teams. The role requires IP law, technical understanding, business and financial skills, negotiation and communication. It may be organised centrally, within business units, or through a mix of central policy and unit level managers.

Legal Compliance

The IA manager must ensure that the statement of working of patents in **Form 27** is filed once for every period of three financial years (Patents Rules, rule 131, as amended in 2024), that renewal fees are paid on time (a 10 per cent reduction applies when fees are paid online four or more years in advance), that assignments and licences of patents are registered (Patents Act, s. 69), that permission is obtained before filing abroad first for inventions made in India by residents (s. 39), and that trade marks are renewed every ten years. Records must also support the auditor's report on intangible assets under CARO 2020.

Part Two: Portfolio Management

2.1 IP Categories and Protection Strategy

IP Categories

Indian law offers a menu of rights, each with its own subject matter, term and formality. A **patent** under the Patents Act, 1970 protects inventions that are new, involve an inventive step and are capable of industrial application, for 20 years from filing, subject to the exclusions in s. 3. A **design** under the Designs Act, 2000 protects the shape, pattern or appearance of an article for 10 years, extendable by 5. A **trade mark** under the Trade Marks Act, 1999 protects signs that distinguish goods or services for renewable periods of 10 years, without limit. **Copyright** under the Copyright Act, 1957 arises automatically in literary, dramatic, musical and artistic works, films, sound recordings and software, generally for the life of the author plus 60 years. **Geographical indications** protect goods linked to a place, and separate laws protect semiconductor layout designs and plant varieties. **Trade secrets** have no statute in India; they are protected through contracts and the equitable duty of confidence for as long as they remain secret.

Meaning of Protection Strategy

A **protection strategy** decides which right to use, when, where and how much, so that IP supports the business at the lowest total cost and risk. The choice depends on the type of innovation, whether rivals can copy it by examining the product, the length of its commercial life, whether infringement can be detected and proved, the cost of filing, renewal and enforcement, the need to avoid disclosure, the markets and competitors involved, and the legal exclusions. In India, for example, s. 3(d) bars patents on new forms of known substances without enhanced efficacy, and s. 3(k) excludes computer programs as such and business methods, which pushes software firms towards copyright, secrecy and contracts.

Patent or Trade Secret

A patent gives a strong exclusive right for 20 years, even against independent inventors, but requires full public disclosure. A trade secret can last indefinitely and needs no disclosure, but it protects only against misuse of confidential information and gives no protection against independent discovery or reverse engineering. Granstrand identifies four situations in which secrecy may be the better choice: when the firm has a large lead over rivals, when the secret is hard to discover (for example a process that leaves no trace in the product), when infringement of a patent would be hard to detect, and when a patent could easily be invented around. He also notes that the legal protection of secrets is weak, so firms must rely on their own measures. In India this

means confidentiality clauses in employment and supply contracts, non disclosure agreements and access controls, because s. 27 of the Indian Contract Act, 1872 makes most post employment non compete restraints void.

Multi Protection, Disclosure and Timing

Granstrand recommends **multi protection**, which means combining several rights around one business: for example a secret formula, patented machinery, a registered design for the container and a registered trade mark for the brand. Trade marks deserve special attention because they can be renewed without limit and grow in value. **Defensive publication** is the deliberate publication of an invention so that nobody can patent it; it is a cheap way of keeping freedom to operate, because published matter forms prior art that defeats later patent claims. Timing is critical: filing must come before any public disclosure, since Indian law allows only limited exceptions for prior disclosure. A first filing gives a priority period of 12 months for patents and 6 months for designs and trade marks to file abroad under the Paris Convention, and the PCT route allows one international application with entry into the Indian national phase at 31 months from priority. A resident inventor must obtain permission before filing abroad first (Patents Act, s. 39). Rights should be sought where the firm sells, where it manufactures and where competitors operate.

2.2 IP Categorisation: Business, Technical and Legal Perspectives

Purpose

Categorisation turns a list of registrations into information for decisions. It shows which rights matter to the business, which only cost money, where the legal risks lie and what action each asset needs.

Three Perspectives

From the **business perspective**, rights are grouped as core or non core, as strategic, defensive or offensive, as revenue earning or merely cost generating, and by product line, business unit or market. A widely used practice sorts patents into those the business is using, those it will use and those it will not use; the last group is licensed, sold, donated or abandoned. From the **technical perspective**, rights are grouped by technology field (IPC or CPC classes), by basic technology or application, by product or process, by basic invention or improvement, and by stage of maturity. From the **legal perspective**, rights are grouped by type, by registered or unregistered status, by legal status (pending, granted, opposed, lapsed), by countries covered and remaining term, by ownership, by licences and security interests already granted, and by strength in terms of validity, claim scope and enforceability. Joint ownership needs particular care, because a joint owner of an Indian patent cannot grant a licence or assign a share without the consent of the other owners (Patents Act, s. 50).

Portfolio Matrix and Decisions

A simple **portfolio matrix** scores each asset for business value and legal strength. Rights high on both are the firm's **crown jewels** and should be maintained, enforced and extended. Rights of high business value but weak legal position form a **weak core** that must be strengthened through further filings, secrecy or better contracts. Rights with a strong legal position but low business value are candidates for **licensing or sale**. Rights weak on both should be **dropped** by abandonment, sale or donation. The firm should list all assets, score them with input from business, technical and legal staff, decide the action for each and repeat the review before renewal fees fall due. Renewal makes sense only where the expected benefit exceeds the cost. Abandonment saves money but may free the technology for competitors, and a lapsed patent can be restored only within the period and conditions allowed by law (Patents Act, s. 60). Firms should also avoid depending on a single patent, licensee or market.

2.3 IP Evaluation and Valuation

Evaluation and Valuation Distinguished

Evaluation is a judgement of the quality and importance of an IP asset, usually expressed through scores or ratings, and it is used for portfolio decisions and screening. **Valuation** fixes a value in money for a stated purpose and date, and it is used in sales, licences, accounts, tax, loans and damages. Evaluation comes first, because a valuation of a weak or invalid right is meaningless. Evaluation looks at legal factors (validity, scope, enforceability, remaining term, territory and ownership), technical factors (novelty, substitutes and ease of inventing around), commercial factors (market size, growth, margins and competition), strategic factors (fit with business plans and blocking power) and risk.

Purposes and Legal Contexts

Valuation is needed for licensing and royalty setting, assignment, mergers and acquisitions, investment, financial reporting, tax and transfer pricing, damages in infringement suits, insolvency proceedings, loans secured on IP and internal portfolio decisions. Several of these have a legal basis: company law requires registered valuers for statutory valuations (Companies Act, 2013, s. 247), tax law requires related party royalties to be at arm's length (Income Tax Act, 2025, ss. 161 to 173), accounting standards require fair values in acquisitions and impairment tests (Ind AS 103, Ind AS 36 and Ind AS 113), and insolvency procedures require estimates of fair value and liquidation value. ICAI Valuation Standard 302 deals specifically with intangible assets.

The Three Approaches

The **cost approach** values an asset at what it would cost to recreate or replace it, less obsolescence. It is simple and suits early stage technology and software, but cost is not the same as value. The **market approach** values an asset by reference to prices or royalty rates in comparable

transactions; it is persuasive when good comparables exist, but true comparables are rare and deal terms are usually confidential. The **income approach** values an asset as the present value of the future income it will generate. Its methods include discounted cash flow, **relief from royalty** (the royalties saved by owning the asset rather than licensing it), **excess earnings** (business profit minus a fair return on all other assets used), **with and without** (the difference in business value with and without the asset) and **premium profit** (extra price or cost saving due to the asset). Real option models, empirical models and index scores complement these approaches.

In practice, the cost approach or real options suit early patents with no income, relief from royalty suits established brands and technologies, excess earnings suits the core technology of a business, the with and without method suits non compete agreements, and comparable royalties combined with discounted cash flow guide licence negotiations. Value is affected by remaining legal life, economic life, exclusivity, claim scope, enforceability, market size, competition, complementary assets, territorial coverage and licences already granted. A sound valuation fixes the purpose, basis and date, identifies the legal rights, evaluates them, collects data, uses at least two methods, tests the sensitivity of the result and states its assumptions clearly.

2.4 IP Value Index

An **IP value index** is a single score that ranks patents or portfolios by quality by combining several indicators. It is a tool for screening and comparison, not for fixing a price, and there is no statutory IP value index in Indian law. Typical indicators include forward citations (later patents citing the patent, a sign of technical importance), family size (the number of countries in which protection is sought, a sign of expected commercial value), the number of claims (breadth), grant lag, backward citations, remaining term, renewals and the history of licensing and litigation.

The best known example is the **Patent Asset Index** of Ernst and Omland, used in commercial patent analytics. Each patent family receives a **Technology Relevance** score based on citations, with an average of 1, and a **Market Coverage** score based on the size of the markets in which it is protected, with a granted United States patent equal to 1.

Competitive Impact = Technology Relevance × Market Coverage; Patent Asset Index = Σ Competitive Impact

If Portfolio A has three families with Competitive Impact of 3.6, 0.15 and 1.08, its index is **4.83**; if Portfolio B has ten families of 0.2 each, its index is **2.0**, so the smaller portfolio is stronger. Other indices include the OECD patent quality index (forward citations, family size, claims and generality, with backward citations and grant lag in an extended version), the Lanjouw and Schankerman composite (which found forward citations most informative) and technology strength measures that multiply patent counts by citation impact. Granstrand describes indirect valuation through indicators such as patent volume, patent shares, citations, renewal fees and the age profile of patents. An index is built by choosing indicators, normalising them, weighting them, combining them and testing the result against real outcomes. Its limits are that it gives no money value, cita-

tions take years to accumulate, weights are subjective, indicators can be manipulated and legal validity is not checked.

2.5 Empirical Valuation Models

Empirical models estimate the value of IP from real data on large numbers of patents and firms rather than from forecasts for a single asset. Granstrand cautions that patent data alone cannot value a portfolio because patent values are highly uneven, and that it must be combined with stock market data, panel data or survey evidence.

The **market value model** relates the stock market value of firms to their tangible assets and knowledge assets:

$V = q \times (A + \gamma K)$

Here V is market value, A is tangible assets, K is the knowledge stock (R&D or patents) and γ shows how the market values knowledge relative to tangible assets. **Renewal models** rely on the fact that owners keep paying renewal fees only while a patent is worth more than the fee, so renewal data reveal how values are spread; the main finding is that most patents are worth little while a few are worth a great deal. **Citation models** show that highly cited patents tend to be more valuable. **Survey models** ask owners to estimate the value of their patents, **event studies** measure share price reactions to patent grants or court outcomes, and **transaction models** use prices from patent sales and licences. These models are useful for policy, screening and benchmarking, and they inform assumptions in individual valuations, but they give averages rather than the value of one patent, depend on available data and show association rather than cause.

2.6 Option Based Valuation Models

A patent gives its owner the **right but not the obligation** to exploit an invention exclusively. Like a call option in finance, it lets the owner wait for information and invest only if conditions are favourable, or expand, license, switch use or abandon. Ordinary discounted cash flow assumes a fixed plan and ignores this flexibility, so it can undervalue early and uncertain IP. In the option analogy, S is the present value of expected cash flows from using the invention, K is the investment needed to commercialise it, T is the time until the decision must be taken (limited by the remaining patent term), σ measures the uncertainty of the project's value and r is the risk free rate. Each renewal fee can also be seen as buying an option to keep the patent for another year.

The **Black and Scholes model** gives the value of such an option:

$C = S \times N(d_1)$ minus $K \times e^{(\text{minus } rT)} \times N(d_2)$

$d_1 = [\ln(S \div K) + (r + \sigma^2 \div 2) \times T] \div (\sigma\sqrt{T})$; $d_2 = d_1$ minus $\sigma\sqrt{T}$

If S is 100, K is 120, T is 3 years, σ is 0.40 and r is 0.07, then d_1 is 0.3864, d_2 is minus 0.3065, $N(d_1)$ is 0.6504 and $N(d_2)$ is 0.3796. The present value of K is 97.27, so the option value is 65.04 minus 36.93, or **28.11**. Investing immediately would have an NPV of minus 20, yet the right to wait is worth 28.11, and the value rises as uncertainty rises.

The **binomial model** assumes that value moves up or down by set factors. With S of 100, an up factor of 1.4, a down factor of 0.7, a risk free rate of 7 per cent and K of 110, the option pays 30 in the up state and nothing in the down state. The risk neutral probability is (1.07 minus 0.7) ÷ (1.4 minus 0.7), or 0.5286, so the option value is 0.5286 × 30 ÷ 1.07, or **14.82**. A **decision tree** gives similar insight: if a pilot costs 20 now, the market next year is good (value 180) or bad (value 40) with equal chance, launch costs 80 and the discount rate is 10 per cent, then committing to launch in both cases gives an NPV of 7.27, while launching only in the good case gives 25.45, so flexibility is worth **18.18**. Option models value flexibility and suit early, risky IP, but the analogy with traded options is imperfect, uncertainty is hard to estimate, rivals may invent around the patent and the patent may be revoked.

2.7 Measuring and Monitoring Intellectual Capital

Measuring IC gives a picture at one point in time, while **monitoring** tracks IC regularly through indicators and targets so that management can act early. Indicators may measure inputs (R&D spending, training hours), processes (projects running, invention disclosures), outputs (patents granted, products launched) or outcomes (revenue from new products, licence income), and they may be leading (predicting the future) or lagging (reporting the past). Human capital is tracked through training, staff turnover and the number of inventors; structural capital through documented processes, IT systems and quality certification; innovation through R&D intensity, disclosures, patents and revenue from new products; and relational capital through customer retention, satisfaction, partners and licensees.

A monitoring system starts from strategy, selects a small number of clearly defined indicators with a named owner and data source, sets targets, reports results in a dashboard or IC statement (the Danish model combines a knowledge narrative, management challenges, initiatives and indicators) and reviews the indicators regularly. Its main problems are poor data, weak comparability between firms, the assumption that indicators cause results, and **gaming**, where people improve the number rather than the reality, for example by filing weak patents to raise patent counts. Start ups monitor milestones and the team, established firms monitor portfolio quality and licence income, universities monitor disclosures, licences and spin offs, and service firms monitor expertise and client relationships.

2.8 Understanding and Managing Knowledge Assets for Competitive Advantage

Data are facts, information is data with meaning, and knowledge is information combined with experience that allows action. **Knowledge assets** are the stocks of useful knowledge a firm holds: know how, skills, processes, designs, software, customer insight and IP. **Tacit knowledge** lives in people's heads and is hard to express, while **explicit knowledge** is written or coded and easy to transfer. Knowledge is not used up when shared, spreads easily, builds on itself and cannot be taken back once disclosed. Arrow's **information paradox** explains why IP rights matter: a buyer cannot judge knowledge without seeing it, but once it is seen it has been given away; patents and confidentiality agreements allow disclosure without loss of protection.

Knowledge gives competitive advantage when it is valuable, rare, hard to imitate and not substitutable (Barney's VRIN test). Prahalad and Hamel's **core competencies** are forms of collective learning that open many markets, add customer value and are hard to copy. Teece's **dynamic capabilities** are the ability to sense opportunities, seize them and reconfigure resources, and Cohen and Levinthal's **absorptive capacity** is the ability to recognise and use outside knowledge. Knowledge management means creating, capturing, storing, sharing, using and protecting knowledge. The **SECI model** of Nonaka and Takeuchi explains how knowledge is created through socialisation (tacit to tacit), externalisation (tacit to explicit), combination (explicit to explicit) and internalisation (explicit to tacit).

Legally, knowledge assets are protected through IP rights, trade secret policies, non disclosure agreements and confidentiality and assignment clauses in employment contracts. Granstrand found that an internal secrecy policy was the most important secrecy measure in large Japanese firms. In India, post employment non compete clauses are generally void under s. 27 of the Indian Contract Act, 1872, so firms rely on confidentiality duties and retention measures such as incentives, career paths and succession planning. Knowledge that includes personal data is also subject to the Digital Personal Data Protection Act, 2023.

2.9 Knowledge Assets in Innovation and Product Development

New product development turns knowledge about customers, technology and markets into products, and firms that reuse knowledge through platforms and standard components innovate faster and at lower cost. IP management must be built into each stage of development. At the idea stage, ideas should be recorded and patent literature searched for existing solutions. At the concept stage, patentability and freedom to operate should be checked. At the design stage, the firm must decide between patenting and secrecy and must file before any disclosure. During testing, non disclosure agreements should bind testers and improvements should be filed. At launch, trade marks and designs should be registered and a final clearance done, and after launch the firm should file improvements, watch competitors and enforce its rights.

Granstrand's guidance fits this process: check others' patents before entering a new field, file early ("speed to patent"), patent continuously through the product life, and use patent maps to generate ideas, track rivals, plan R&D and strengthen bargaining. He reports that Toshiba reviews patents at every step of the innovation process. The legal points that shape development in India are that public disclosure before filing usually destroys novelty, that the exclusions in s. 3(d) and s. 3(k) affect pharmaceutical and software inventions, that residents need permission before filing abroad first (s. 39), and that joint development requires clear agreement on ownership because of s. 50.

2.10 Patent Stocks

A **patent stock** is the accumulated number of patents of a firm, industry or country, used as a measure of its stock of technical knowledge. It can be counted as a simple number of applications or grants, as patent families (one invention counted once across countries), as active patents still in force, as a citation weighted count, as a quality adjusted count, or as a depreciated economic stock. The depreciated stock gives less weight to older patents because knowledge becomes outdated:

$K_t = (1 \text{ minus } \delta) \times K_{t \text{ minus } 1} + P_t$

Here K is the stock, P is new patents in the year and δ is the yearly rate of decline, often taken as about 15 per cent. If a firm obtains 10, 12, 15, 9, 20, 18 and 22 patents from 2019 to 2025, the simple count is 106, but the depreciated stock in 2025 is only **74.20**.

Patent stocks also support the measure of **revealed technological advantage**:

RTA = (Firm's patents in a field ÷ Firm's total patents) ÷ (World patents in the field ÷ World total patents)

If 20 per cent of a firm's patents are in a field that has 5 per cent of world patents, its RTA is 4.0, showing strong specialisation. Patent stocks are used to measure innovation, explain firm value, compare competitors and screen acquisition targets. Their limits are that industries patent at different rates, patent values are highly uneven, strategic filing inflates counts, publication is delayed by about 18 months and company names vary. Granstrand insists that the counting unit must always be stated.

2.11 Copyright Collective Management

Collective management is the system in which **copyright societies** license works on behalf of many owners, collect royalties and distribute them. Users such as radio and television broadcasters, hotels, restaurants, event organisers and digital platforms use thousands of works, and negotiating separately with every owner would be impossibly costly, while individual owners cannot monitor use across the country. A society therefore grants **blanket licences** covering its whole

repertoire, sets tariffs, collects royalties, tracks use, distributes income to members, enforces rights and makes reciprocal arrangements with foreign societies. Societies commonly manage performing rights in music and lyrics, rights in sound recordings, performers' rights, reprographic rights and rights in scripts. Collective management may be voluntary, extended to non members by law (as in Nordic countries), or mandatory for certain rights.

Because one society tends to dominate each class of works, collective management raises concerns about high tariffs, unfair treatment of users, unclear distribution and high costs, which is why the law regulates it. Under s. 33 of the Copyright Act, 1957, the business of issuing or granting licences in works may be carried on only by or through a registered copyright society, although owners keep the right to license their own works individually; ordinarily only one society is registered for each class of works, and registration lasts five years. Section 33A requires each society to publish its tariff scheme, s. 34 governs the administration of rights entrusted to the society, s. 35 gives owners and authors control over the society, and s. 36 requires returns and audited accounts. The Copyright Rules, 2013 add detailed procedures. The 2012 amendments required societies to register afresh, strengthened the control of authors and gave authors of works used in films and sound recordings a right to share in royalties. Organisations active in this field in India include IPRS for music and lyrics, PPL and RMPL for sound recordings, ISRA for performers and IRRO for reprographic rights; current registration status should be checked with the Copyright Office. Internationally, CISAC links authors' societies, and the European Union regulates the governance and transparency of collective management through Directive 2014/26/EU.

2.12 The Role of Intellectual Capital in Valuing Knowledge Companies

Knowledge companies, such as software, pharmaceutical, media and platform businesses, derive most of their value from technology, data, brands, people and relationships; Granstrand calls them IC firms. Their financial statements understate their value because research and brand building are expensed under Ind AS 38, employees are not shown as assets, assets are carried at historical cost, and network effects, data and future opportunities are not captured. Their value is driven by IP, key people, data, software, research pipelines, brands, customer bases, processes, growth and scalability, together with the risks attached to each.

It is important to separate different measures of value. **Enterprise value** is the value of the business to all providers of capital and equals equity value plus net debt; **equity value** is the value to shareholders; **asset value** is the value of individual assets; **IP value** is the value of the legal rights and the income they protect; and **IC indicators** are non money measures of intellectual capital. Knowledge companies are usually valued by discounted cash flow with scenarios, by market multiples such as enterprise value to revenue, by adding up the values of their parts, by real options for research pipelines and by customer lifetime value, sometimes after treating R&D as an investment rather than an expense.

The **residual method**, which treats market value minus book value as the value of intellectual capital, is unreliable. The gap includes market mood and growth expectations, ignores undervalued tangible assets and hidden liabilities, and changes every day. If market value is ₹10,000 crore and book value ₹2,000 crore, the gap is ₹8,000 crore; if the share price then falls by 20 per cent, the gap becomes ₹6,000 crore although nothing inside the firm has changed.

Part Three: Innovation and IP Protection

3.1 Innovation and Market

Invention is a new idea or device, **innovation** is the successful introduction of a new or improved product, process, service or business model into the market, and **diffusion** is its spread among users. Schumpeter described innovation as "new combinations" of goods, methods, markets, sources of supply and forms of organisation, and he described the resulting "creative destruction", in which new products and firms replace old ones. Innovation may be classified by its object (product, process, service, business model or organisation), by its degree (incremental or radical), by its market effect (sustaining or disruptive) and by its architecture. Henderson and Clark distinguish **incremental** innovation (core ideas and links between components unchanged), **modular** innovation (core ideas replaced, links unchanged), **architectural** innovation (core ideas kept, links changed) and **radical** innovation (both changed).

Innovation arises from **technology push**, where new science creates products, and **market pull**, where customer needs drive development. Schumpeter argued that large firms with market power innovate more, while Arrow argued that competition gives stronger incentives; in practice the relationship varies by industry. Rogers explains diffusion through adopter groups: innovators (2.5 per cent), early adopters (13.5 per cent), early majority (34 per cent), late majority (34 per cent) and laggards (16 per cent). A technology usually follows an **S curve**, improving slowly, then rapidly, then levelling off, and markets often settle on a **dominant design** after which competition shifts to price and process. In networked markets, products become more valuable as more people use them, and patents included in standards earn royalties from all implementers. Granstrand observes that firms strong in an old technology are often slow to build patent positions in a new one, which gives entrants an opening, and that old technologies may improve under threat, the "sailing ship effect".

IP links innovation to the market. It allows innovators to capture returns, makes technology tradeable through licences and assignments, signals capability to investors and supports financing. Granstrand places IP strategy on a "strategy ladder" below the firm's business, innovation, technology and commercialisation strategies, which means that IP choices should follow business choices.

3.2 Radical Innovation

A **radical innovation** is a major technical change that creates a new technology base, a large jump in performance or an entirely new market, as when transistors replaced vacuum tubes or

digital technology replaced analogue mobile telephony. It involves high uncertainty and cost, long development, the obsolescence of existing skills and frequent entry by newcomers, and it often creates new industries. For IP management, the innovator should file early with broad claims, seek **strategic patents** that rivals cannot easily invent around, and quickly file follow up patents, since Granstrand describes a second race for **surrounding patents** around a key patent. Broad claims attract opposition and revocation attempts, so drafting must withstand the inventive step requirement in s. 2(1)(ja) of the Patents Act. Processes may be protected as trade secrets and the new brand by trade marks. Radical projects need patient finance such as venture capital, government grants or staged corporate investment, and they face risks of technical failure, slow adoption, lack of complementary assets and regulatory delay.

3.3 Disruptive Innovation

Disruptive innovation, as explained by Christensen, is a process by which a smaller firm with fewer resources successfully challenges established firms. Established firms improve their products for their most demanding and profitable customers, which is sustaining innovation, and in doing so overshoot the needs of other customers. The entrant starts either at the **low end**, serving customers the leaders neglect, or in a **new market**, serving people who were not buying at all, with a simpler, cheaper or more convenient product that is "good enough". It then improves and moves into the main market, and disruption is complete when mainstream customers adopt it. Classic examples include steel minimills against integrated mills, personal computers against minicomputers, and Netflix against video rental chains. Christensen and his co authors argue that Uber was not disruptive, because it began in the main market with a better service.

Disruption should not be confused with mere technical novelty: a disruptive product is usually worse at first on conventional measures but cheaper or easier to use, and leaders ignore it because its margins are low. For IP management, disrupters rely more on business models, speed and network effects than on patents, and business methods as such are not patentable in India (s. 3(k)), so their protection comes mainly from copyright in software, trade secrets, brands and contracts. Established firms' patent portfolios, built for sustaining innovation, may not block a new business model, so incumbents must watch low end and new market entrants, create separate units, or invest in or acquire the newcomers.

3.4 Sustaining and Sustainable Innovation

The syllabus term may refer to **sustaining innovation** in Christensen's sense or to **environmentally sustainable innovation**, and an answer should state which meaning it uses; beside radical, disruptive and incremental innovation, sustaining innovation is the likely meaning. Sustaining innovation improves existing products for existing customers, whether by small steps or major breakthroughs, and established firms usually win these contests because they have the resources and motivation. The matching IP strategy is continuous follow up patenting, large portfo-

lios, cross licensing and protection of designs and brands. **Sustainable (green) innovation** reduces environmental harm through cleaner energy, recycling or low emission processes, and it is driven by regulation, consumers, cost savings and investors. Its IP questions concern diffusion: green patents attract investment, wide licensing can spread clean technology, platforms such as WIPO GREEN connect owners and users, and technology transfer to developing countries takes place through licences and partnerships.

3.5 Incremental Innovation

Incremental innovation consists of small, continuous improvements to existing products or processes, such as better features, lower cost or higher quality. It is low in risk, builds on existing knowledge and keeps firms competitive. Granstrand shows that firms protect incremental innovation through **continuous patenting** throughout the product life, and that minor patents, which engineers sometimes dismiss as "petty" or "junk", can still be valuable as bargaining chips, although too many weak patents waste money. Firms also use fencing and surrounding strategies, register designs for changes in appearance and keep process improvements secret. Indian law sets limits: obvious changes fail the inventive step test, s. 3(d) requires enhanced efficacy for new forms of known substances, India has no utility model or petty patent system, and the law guards against "evergreening", meaning the extension of protection through minor changes.

Comparing the four types: radical innovation creates a new technology base and new markets and calls for pioneer and strategic patents; disruptive innovation uses a simple, cheap, good enough offering through which newcomers displace leaders and relies on business model and speed; sustaining innovation improves products for current customers, strengthens leaders and relies on follow up patents; and incremental innovation makes small gains and relies on continuous patents, designs and secrets.

3.6 Open Innovation and Its Principal Models

Open innovation, a term developed by Chesbrough, means using ideas from outside the firm as well as from inside, and using external as well as internal routes to market. Closed innovation assumes that the best people work for the firm, that the firm must invent, develop and sell everything itself, that being first to market wins and that IP should be controlled so that nobody else profits. Open innovation assumes instead that many capable people work elsewhere, that external R&D can create value, that a better business model matters more than being first, and that a firm should earn from others' use of its IP and buy others' IP when it helps its business.

Gassmann and Enkel identify three models. In the **inbound (outside in)** model, the firm brings in external knowledge through licensing in, acquiring start ups, crowdsourcing and working with suppliers and users. In the **outbound (inside out)** model, it takes internal ideas out through licensing, sale of IP and spin offs. In the **coupled** model, it creates knowledge jointly with partners through alliances, consortia and joint development. Common forms include collaborative re-

search, university and industry projects, corporate venture capital, open source development, innovation contests and ecosystems built around platforms. Granstrand's earlier work already classified the ways firms acquire technology (own R&D, buying firms, joint ventures, buying technology and scanning) and exploit it (own use, new units, joint ventures and selling or licensing technology). Bogers, Bekkers and Granstrand describe four ways of sharing knowledge within a collaboration: open to all, open within the group only, **layered** (full sharing among core members and limited sharing with outer members) and closed. Open innovation brings more ideas, lower cost and risk, faster launch and income from unused IP, but it also brings loss of control, leakage of knowledge, the "not invented here" attitude, coordination costs and disputes over ownership.

3.7 Open Innovation and IP

Open innovation appears to conflict with IP, because firms must share knowledge to collaborate but must protect it to profit. In fact, IP makes openness possible: clear ownership allows firms to disclose knowledge safely and to license it on agreed terms. Every collaboration should identify four kinds of knowledge: **background IP** brought in by each party, **foreground IP** created in the project, **sideground IP** developed in house in parallel with the project and **postground IP** developed after it ends. The collaboration agreement must settle the use of background IP, the ownership of foreground IP (sole or joint), access rights during and after the project, rights in improvements and any grant back, confidentiality, publication rights (important for universities), licensing terms, data rights, decision making, dispute resolution and exit.

Licences are the building blocks of collaboration. An **exclusive** licence excludes even the licensor, a **sole** licence allows only the licensor and licensee to use the IP, and a **non exclusive** licence allows many licensees. Firms also use cross licences, sublicences, blanket licences covering future developments, package licences, grant backs and patent pools. In software, **permissive** open source licences (such as MIT, BSD and Apache) allow use in closed products, while **copyleft** licences (such as the GPL) require modified versions to remain open, so firms must track the open source code they use.

Indian law shapes these arrangements. A joint owner of a patent cannot license it or assign a share without the consent of the other owners (Patents Act, s. 50), so agreements must deal with joint ownership expressly. Patent licences must be in writing and registered (ss. 68 and 69). Section 140 makes void certain restrictive conditions in patent licences, including tying arrangements, exclusive grant backs, bars on challenging validity and coercive package licensing, and s. 3(5)(i) of the Competition Act, 2002 protects only reasonable conditions needed to protect IP rights. India has no statute equivalent to the United States Bayh Dole Act on publicly funded research, so universities and research institutions rely on their own IP policies and contracts.

3.8 Patent Databases as a Source of IP Management Information

Patent documents disclose technology in a standard format, often before it appears anywhere else. Granstrand calls patent information a "tremendous and unique source of technical knowledge" and found that learning from patent documents was the most important method of tracking others' technology in large Japanese firms. A patent document contains bibliographic data (numbers, dates, owners, inventors and classification), an abstract, a description, claims that define the legal scope, citations, family information and legal status.

For IP management, patent databases support prior art searches before filing, patentability assessment, technology landscaping, competitor intelligence and monitoring, freedom to operate checks, identification of licensees, partners and acquisition targets, discovery of expired patents that are free to use, analysis of patent families and legal status, citation analysis, trend analysis and portfolio benchmarking. Indian users rely on the Indian Patent Office's search system and electronic register, Espacenet of the European Patent Office, PATENTSCOPE of WIPO, the search tools of the United States Patent and Trademark Office, free services such as Google Patents and The Lens, and paid analytics databases. Searches combine classification codes (IPC and CPC), keywords joined by AND, OR and NOT, applicant and inventor names, number searches, family searches, citation searches and date limits, and legal status should always be confirmed on the official register.

Different purposes call for different searches: a **novelty or patentability search** asks whether an invention can be patented, a **freedom to operate search** asks whether a product infringes live patents, a **validity search** asks whether a patent can be revoked, a **landscape search** shows who is active in a field, and a **due diligence search** checks ownership, status and risk in a transaction. **Patent mapping** turns results into charts of technologies, owners, trends and gaps; Granstrand notes that it began at the Japan Patent Office, which published its first patent map in 1968. Users must remember that coverage and data quality differ between databases, machine translations can mislead, applications are published only after about 18 months, company names vary, legal status data may be outdated and raw counts can mislead.

3.9 IP Due Diligence

IP due diligence is a careful legal and commercial investigation of IP before a transaction. It answers four questions: whether the target owns or validly controls the IP, whether the IP is valid and enforceable, whether the business is free to operate without infringing others' rights, and what the IP is worth after taking risks into account. It is carried out in mergers and acquisitions, investments, licensing, technology transfer, joint ventures, loans, purchases of distressed assets in insolvency and start up funding.

The process begins with planning the scope and team, followed by a questionnaire and document request, review of documents in a data room, independent searches of official registers

and charge records, interviews with managers and inventors, analysis and a report on risks. The findings are then reflected in the deal through price adjustment, warranties, indemnities, conditions to be met before completion and escrow, and after completion the transfers must be registered.

The investigation covers ownership and the **chain of title**, which requires written and registered assignments from inventors and earlier owners (Patents Act, ss. 68 and 69; Copyright Act, s. 19; Trade Marks Act, s. 45). It checks registration and renewal status, including filing of Form 27, and searches for **encumbrances** such as charges registered under s. 77 of the Companies Act, 2013 and security interests filed with CERSAI. It reviews licences given and taken, especially exclusivity and change of control clauses, IP and confidentiality clauses in employment and consultancy contracts (because consultants own their work unless they assign it), trade secret protection, open source licences, infringement notices, pending disputes and oppositions, freedom to operate, territorial coverage and validity risks such as prior art and the exclusions in s. 3. It also covers tax (withholding on royalties, GST and transfer pricing), competition law, personal data and security, dependence on one licence or customer, the link between revenue and IP, and integration after the deal. Due diligence differs from an **IP audit**, which is an internal and regular review not tied to a deal, from a **valuation**, which produces a money figure, from a **legal opinion**, which answers a specific legal question, and from a **freedom to operate analysis**, which examines infringement risk for a particular product.

3.10 Drafting a Licence Agreement

A **licence** is permission from the owner (the licensor) to another person (the licensee) to do acts that would otherwise infringe the owner's IP, on agreed terms. Ownership remains with the licensor, whereas an **assignment** transfers ownership. Before drafting, the parties must be clear about their business aims, complete due diligence on title, validity and freedom to operate, agree the value and royalty, define the scope of the grant, satisfy legal formalities, respect competition law, plan for tax and foreign exchange, arrange regulatory approvals, provide for quality control and plan the exit.

Indian law imposes specific requirements. A patent licence must be in writing and registered (Patents Act, ss. 68 and 69), a joint owner cannot license without the consent of the others (s. 50), and the restrictive conditions listed in s. 140 are void. A copyright licence must be in writing (Copyright Act, s. 30), and under s. 30A the requirements of s. 19 apply, so the licence should identify the work, the rights, the duration, the territory and the royalty. A trade mark licence must preserve the licensor's control over quality, so that the mark continues to indicate a single trade source. Licence terms are also subject to s. 3(5)(i) of the Competition Act, 2002, and royalty payments attract tax deduction at source, GST at 18 per cent and treaty relief for cross border payments.

Royalties may be structured as a lump sum, a running royalty calculated as a percentage of net sales or an amount per unit, an upfront fee combined with a running royalty, a minimum annual royalty, milestone payments, step down rates, equity or a cross licence, and **net sales** must be defined carefully.

The agreement should identify the parties and their affiliates and define the licensed patents, products, net sales, territory and field. The **grant clauses** state the licensed rights, confirm the licensor's ownership, define the scope (make, use, sell or import), the type of exclusivity, the territory, the field of use, the duration and any right to sublicense. The **financial clauses** set out the consideration, royalty rate and base, minimum guarantees, milestones, reports, records and audit rights, and the allocation of taxes, including deduction at source and grossing up. The **operational clauses** cover quality control, technical assistance, regulatory approvals and registration of the licence. The **IP clauses** decide who owns improvements and new IP, avoiding exclusive grant backs, and protect confidential know how. The **risk clauses** contain representations, warranties on title (validity is usually not warranted), indemnities, and rules on who sues infringers and who defends claims against the licensee. The **exit and legal clauses** deal with termination for breach, non payment or insolvency, the consequences of termination (stopping use, selling off stock, returning materials and dealing with sublicences), dispute resolution, governing law, cross border issues such as foreign exchange and withholding tax, force majeure, assignment and change of control, and data, software and source code escrow. These clauses work together: exclusivity is usually paired with a minimum royalty, field and territory limits must match the improvement clauses, termination must address sublicences and stock, and warranties and indemnities must be read together.

3.11 IP Valuation and Finance

IP valuation is used in licensing, sale and purchase, investment, financial reporting, taxation, damages, insolvency, portfolio decisions and lending. Several related terms must be kept apart. **Value** is an estimate of worth for a purpose and date, **price** is the amount actually agreed, **consideration** is what is actually given (cash, shares or a cross licence), **carrying amount** is the figure in the accounts and **collateral value** is a lender's careful estimate of what it could recover on default.

An IP right also has more than one life. Its **legal life** is the term of protection, its **economic life** is the period during which it earns income and is usually shorter, and its **useful life** is the accounting estimate used for amortisation. Economic life is shortened by new technology, slow market adoption, weak enforcement, competition and regulation. Valuers handle this uncertainty through scenarios, sensitivity analysis, probability weighting, simulation and real options.

IP can be financed in several ways: loans secured on IP, the sale of future royalties to investors, securitisation of royalty streams (an early example being bonds backed by David Bowie's music

royalties in 1997), sale and licence back, venture debt and equity, and government schemes that share lenders' risks, such as Singapore's IP financing scheme.

3.12 Net Present Value Calculations

Money received today is worth more than the same amount received later, so future cash flows are **discounted** to their present value.

$PV = FV \div (1 + r)^t$; $NPV = \Sigma\ [CF_t \div (1 + r)^t]$ minus initial investment

A project or licence is worthwhile if its NPV is positive. The **internal rate of return** is the discount rate at which NPV is zero, and for an asset with no fixed end the **terminal value** is the next year's cash flow divided by (r minus g), where g is the long term growth rate; a patent with a fixed term should not be given a terminal value. The discount rate is often based on the **CAPM** (cost of equity equals the risk free rate plus beta times the market risk premium) or the **WACC** (the weighted average of the cost of equity and the after tax cost of debt), with an extra premium for IP risks such as invalidity and obsolescence. With a risk free rate of 7 per cent, beta of 1.3 and market premium of 7 per cent, the cost of equity is 16.1 per cent; with 60 per cent equity and 40 per cent debt at 10 per cent before 25 per cent tax, the WACC is 0.6 × 16.1 plus 0.4 × 7.5, or **12.66 per cent**.

Example 1: commercialising a patent requires ₹50 crore now and brings ₹15, 20, 25 and 20 crore over four years. At 12 per cent, the discount factors are 0.8929, 0.7972, 0.7118 and 0.6355, so the present values are 13.39, 15.94, 17.79 and 12.71, a total of 59.84. The NPV is 59.84 minus 50, or **₹9.84 crore**, and the IRR is about 20.5 per cent, so the project should go ahead.

Example 2: a licensee expects sales of ₹40 crore growing at 10 per cent a year for five years, the royalty is 5 per cent, the licensor's cost is ₹0.2 crore a year, tax is 25 per cent and the discount rate is 14 per cent. The after tax royalties are 1.35, 1.50, 1.665, 1.8465 and 2.0462, with a present value of **₹5.62 crore**. A lump sum offer of ₹6 crore is worth only ₹4.5 crore after tax, so the running royalty is better unless the licensee's credit or the patent's validity is doubtful. At a 12 per cent discount rate the value rises to ₹5.92 crore, and at 16 per cent it falls to ₹5.34 crore, which shows how sensitive the result is to the rate.

Example 3: for a drug candidate, each cash flow is multiplied by the probability of reaching it. A trial costing 20 now (certain), a further cost of 60 in year 1 (35 per cent chance), a filing cost of 5 in year 3 (21 per cent chance) and income of 90 a year in years 4 to 8 (18.9 per cent chance), discounted at 12 per cent, give present values of minus 20, minus 18.75, minus 0.747 and plus 43.645, so the **risk adjusted NPV is 4.15**; without probabilities the NPV would be a misleading 153.79. NPV depends on forecasts and the discount rate and ignores flexibility, and legal facts such as remaining term, validity and exclusivity must decide the period and risk of the cash flows.

3.13 Benchmarking Methods

Benchmarking compares an IP asset, royalty rate or portfolio with similar transactions or competitors in order to estimate value or judge performance. **Royalty benchmarking** takes the royalty from comparable licences, adjusted for differences, while **industry norms** give typical ranges. The **25 per cent rule of thumb** gives the licensor about one quarter of the licensee's expected profit, so a 20 per cent profit margin suggests a royalty of about 5 per cent of sales; it is simple but widely criticised as arbitrary. **Profit split** divides profit according to each party's contribution, **portfolio benchmarking** compares patent portfolios with competitors using counts and quality indices, and **practice benchmarking** compares IP management practices with leading firms.

Royalty benchmarking defines the asset and rights, searches for comparable licences, checks comparability (industry, exclusivity, territory, field, stage of development, remaining term, strength of protection, payment terms, support included, relationship of the parties and date), adjusts for differences, chooses a range and a point, and cross checks the result with income methods. For example, if comparable licences show royalties of 3.5, 4.0, 5.5, 6.0 and 8.0 per cent, the median is 5.5 per cent and the middle range is 4.0 to 6.0 per cent; an exclusive licence of proven technology without technical support might be priced near 6 per cent, while the 8 per cent licence is less comparable because it included know how and support. Benchmarking is limited by the scarcity of true comparables, the confidentiality of licence terms and the subjectivity of adjustments, and Granstrand warns that patent and R&D figures must refer to the same group of companies before they are compared. In Indian tax law, royalties between related companies must be at arm's length (Income Tax Act, 2025, ss. 161 to 173), which makes comparable licence analysis a routine exercise.

3.14 Discounted Valuation Methods (Discounted Type)

Discounted methods value IP as the present value of its future benefits and together form the income approach. The **relief from royalty** method values an asset by the after tax royalties its owner avoids by not having to license it:

Value = Σ [Revenue × Royalty rate × (1 minus tax rate) ÷ $(1 + r)^t$]

If revenue is 100 and grows by 10 per cent a year for five years, the royalty rate is 4 per cent, tax is 25 per cent and the discount rate is 16 per cent, the after tax royalties saved are 3.00, 3.30, 3.63, 3.99 and 4.39, and their present value is **11.66**. This method suits trade marks and technologies for which royalty rates can be observed.

The **multi period excess earnings method** values the main asset of a business by taking the profit it generates, deducting a fair charge for all other assets used (working capital, machinery, workforce and brand) and discounting the remaining excess earnings. If revenue is 200, 180 and 150, after tax profit is 18.75 per cent of revenue and charges for other assets are 8.5 per cent of

revenue, the excess earnings are 20.5, 18.45 and 15.375, and at 18 per cent their present value is **39.98**. The **with and without** method values an asset as the difference between the business's cash flows with and without it; if a non compete agreement protects cash flows of 50, 55 and 60 that would otherwise fall to 42, 46 and 50, the differences of 8, 9 and 10 discounted at 15 per cent give **20.34**. The **premium profit** method values the extra price or cost saving that a brand or patent allows, and risk adjusted NPV applies probabilities to staged projects. Common mistakes in discounted methods are counting the same income twice, mixing before tax cash flows with after tax rates, giving a patent an unlimited life and borrowing royalty rates from unlike deals.

3.15 Patent Stocks in Valuation and Finance

Patent stocks are used for quick screening of firms, for explaining market values, for a first view of a target's technology in mergers, for portfolio licensing and for benchmarking, and lenders prefer several strong patent families to a single patent. Simple counts can mislead, however, because a few patents carry most of the value, counting units differ (applications, grants or families), flooding and decoy filings inflate numbers, many patents are unused or weak, and some may have lapsed or face challenge. A better approach counts families and active patents, adjusts for quality, depreciates older patents, identifies the most valuable "crown jewel" patents for separate valuation by income methods, and confirms ownership and validity through due diligence. Thus Portfolio A, with three strong families and an index of 4.83, is stronger than Portfolio B, with ten weak families and an index of 2.0, even though B is larger.

3.16 IP as Collateral

Using IP as **collateral** means giving a lender a security interest over IP rights, and often over the royalties they produce, so that the lender can recover the loan from them if the borrower defaults. The borrower usually keeps ownership and use, but the lender may restrict licensing and transfer. Security may take the form of a mortgage of a patent, a charge or hypothecation over IP and royalties, an assignment by way of security with a licence back, or a pledge of shares in a company that holds the IP.

Indian law supports such security. The **SARFAESI Act, 2002** defines property to include intangible assets such as know how, patents, copyrights, trade marks, licences and franchises (s. 2(1)(t)), and allows secured creditors to enforce their security without going to court (s. 13). Security interests over intangible assets must be registered with **CERSAI**, a mortgage of a patent must be in writing and registered under ss. 68 and 69 of the Patents Act, and a company must register charges on its assets with the Registrar of Companies under s. 77 of the Companies Act, 2013. The moratorium under the Insolvency and Bankruptcy Code, 2016 stops enforcement during insolvency resolution. The National IPR Policy, 2016 supports IP based financing, but as on 7

November 2026 there is no dedicated RBI framework for IP backed lending, and such lending remains limited in practice.

A lender must identify the exact rights, verify ownership and the chain of title, value the IP carefully using a distressed value with a **haircut** and a limited **loan to value ratio**, create the security by a written deed, register it in every relevant registry, check priority against earlier charges, monitor renewals, licences and disputes, define default and enforcement, plan for insolvency and cover future IP and royalty income. Lenders worry about uncertain values, the thin market for resale, obsolescence, territorial limits, licence restrictions and invalidity, while borrowers worry about losing the freedom to license and about strict covenants. For example, if a technology is valued at ₹3.63 crore, a 50 per cent haircut gives a distressed value of ₹1.81 crore, and lending half of that gives a loan of only about ₹0.91 crore, which shows that IP alone rarely supports a large loan. Singapore has run a scheme in which the government shared default risk with banks, and WIPO's country reports show that valuation, weak resale markets and lenders' unfamiliarity with IP are common problems.

3.17 IP Audit

An **IP audit** is a systematic review of a firm's IP to find out what it owns or uses, confirm ownership and protection, assess how the IP is used and what it is worth, and identify legal and commercial risks. A **general audit** covers the whole portfolio, an **event based audit** is done before a merger, investment, public issue, loan or product launch, a **focused audit** examines one area such as trade secrets or software licences, a **royalty audit** checks a licensee's payments and a **compliance audit** checks filings and legal duties.

The audit is planned by fixing its scope and team. The auditors then identify all IP through registers, interviews and records, verify ownership and assignments, check the status, deadlines and territorial coverage of protection, review licences and other contracts, check compliance with renewals, Form 27 and the record keeping that auditors examine under CARO 2020, review how the IP is used commercially, assess risks, report their findings, fix the problems found and set up regular monitoring. Typical questions are whether a complete IP register exists, whether assignments are signed and registered, whether deadlines are tracked and fees paid, whether licences match the register, whether employment contracts assign IP and require confidentiality, whether trade secrets are protected by access controls and non disclosure agreements, whether open source licences are respected, whether disputes are tracked and which rights earn money and which only cost money.

Risks are commonly rated by multiplying likelihood (1 to 5) by impact (1 to 5): a score of 15 to 25 is high, 8 to 14 is medium and 1 to 7 is low. The audit report usually contains a summary, the scope, the list of IP, ownership issues, gaps in protection, contract issues, compliance findings, commercial opportunities, a risk table and an action plan. An IP audit differs from due diligence, which serves a particular deal, from valuation, which produces a money figure, from a financial

audit, which tests the accounts, and from a portfolio review, which ranks assets for keep or drop decisions. Its benefits are clear knowledge of the firm's assets, cost savings, new licensing income, reduced legal risk and readiness for transactions and loans.

3.18 IP Taxation

The **Income Tax Act, 2025** applies from **1 April 2026** in place of the Income Tax Act, 1961, and it uses a single "tax year" equal to the financial year; GST applies under the CGST and IGST Acts, 2017. Section 9 defines **royalty** to include payment for the transfer or licence of rights in patents, inventions, designs, secret formulas or processes, trade marks or similar property, for imparting technical or commercial knowledge and for rights in copyright works, whether or not the payer possesses, uses or locates the right in India. Licence fees and royalties are revenue income, an outright sale of IP may give rise to capital gains (or business income for a dealer), and payments for technical help may be fees for technical services. Payment for a copy of software for the buyer's own use, without any transfer of copyright rights, is generally not treated as royalty under tax treaties, whereas payment for the right to copy or modify software is royalty.

For payments to residents, s. 393(1) requires tax to be deducted at source on royalty at 10 per cent (2 per cent for royalty on films and for fees for technical services) where payments exceed ₹50,000 in a year. Royalty earned by non residents is taxed at 20 per cent on the gross amount under s. 207, or at a lower treaty rate where the treaty is more beneficial (s. 159), and the payer must deduct tax under s. 393(2); royalty connected with a permanent establishment in India is taxed as business income. Under s. 33(1)(b), depreciation at 25 per cent is allowed on acquired know how, patents, copyrights, trade marks, licences and franchises, but not on goodwill. The **patent box** in s. 194(1) taxes royalty from a patent developed and registered in India at 10 per cent of the gross amount without any deduction of expenses; it is available to a resident patentee who is the true and first inventor, where at least 75 per cent of the expenditure on the invention was incurred in India, and the option is exercised in Form 65 by the due date of the return. If the patentee departs from the regime in any of the following five years, it loses the benefit for the next five years.

Royalties between related companies must be at arm's length under ss. 161 to 173, and the profit from IP should be allocated to the group company that performs the development, enhancement, maintenance, protection and exploitation (DEMPE) functions. Under GST, a temporary transfer of an IP right or permission to use it is a supply of services (CGST Act, Schedule II, para 5(c)), taxed at **18 per cent**; imports of IP services are taxed under reverse charge, and exports can be zero rated if the conditions for export of services are met. Anti avoidance rules prevent shifting IP income to low tax companies without real activity, and assignments attract stamp duty under state laws. The 1961 Act allowed deductions for scientific research expenditure (s. 35), and the corresponding provision of the 2025 Act should be checked before relying on it.

Quick Revision

Part One: Measuring and Extracting Value

Intangible assets are non physical resources that bring future benefits, and under Ind AS 38 they must be identifiable, controlled and expected to give economic benefits. Because research is expensed and internally generated brands and goodwill are never recognised, accounts show few intangibles, so management relies on wider measurement through identification, classification, indicators and valuation. Public policy balances relevance against reliability; India combines strict recognition (Ind AS 38 and AS 26) with registered valuers, ICAI VS 302, CARO 2020 reporting, depreciation on acquired intangibles, a 10 per cent patent box, SARFAESI and CERSAI security and the National IPR Policy. Intellectual capital consists of human, structural and relational capital and is measured through the Balanced Scorecard, Skandia Navigator, Intangible Assets Monitor, Tobin's q, VAIC and CIV. Profits are extracted through own use, licensing, assignment, collaboration, finance, enforcement, standards and collective licensing, within the limits of s. 140 of the Patents Act, s. 3(5)(i) of the Competition Act and the working requirement behind compulsory licensing. Start ups are measured by team, ownership, filings, freedom to operate and traction and valued by scorecard and venture capital methods. A portfolio is a competitive tool through protection, bargaining and retaliatory power, but its value depends on quality rather than size. The intellectual asset manager has moved from legal administration to business strategy and must keep the firm compliant with Form 27, renewals and registration.

Part Two: Portfolio Management

Protection strategy chooses among patents, designs, marks, copyright and trade secrets according to visibility, life, cost and enforceability, and Granstrand's four situations explain when secrecy is better. Portfolios are categorised from business, technical and legal perspectives and managed through a matrix of crown jewels, weak core, licensing candidates and assets to drop. Evaluation judges quality before valuation fixes money value by the cost, market or income approach. An IP value index such as the Patent Asset Index ranks portfolios but does not price them, empirical models show that patent values are highly uneven, and option models value the flexibility a patent gives. Intellectual capital is monitored through a few clear indicators, with care to avoid gaming. Knowledge assets give advantage when they are valuable and hard to imitate, are created through the SECI process and are protected by contracts and IP within the limits of s. 27 of the Contract Act. Patent stocks measure accumulated knowledge and must state their counting unit, copyright societies are regulated by ss. 33 to 36A of the Copyright Act, and

knowledge companies must be valued from cash flows rather than from the gap between market and book value.

Part Three: Innovation and IP Protection

Innovation may be radical, disruptive, sustaining or incremental, and each type needs a different IP strategy; disruption is a process that begins at the low end or in a new market. Open innovation uses inbound, outbound and coupled models and depends on clear rules for background and foreground IP, joint ownership under s. 50 and lawful licence terms. Patent databases support prior art, freedom to operate, landscaping and competitor intelligence. Due diligence checks title, validity, freedom to operate, encumbrances, contracts and tax before a deal, and a licence must be in writing, registered and carefully drafted on scope, payment, quality, IP ownership, risk and exit. Valuation and finance rely on NPV, benchmarking and discounted methods such as relief from royalty and excess earnings, and patent counts must be adjusted for quality. IP can be offered as collateral under SARFAESI with CERSAI registration, but lenders apply heavy haircuts. An IP audit reviews the whole portfolio regularly. Under the Income Tax Act, 2025, royalty attracts 10 per cent deduction at source for residents, 20 per cent or the treaty rate for non residents, 25 per cent depreciation on acquired intangibles, a 10 per cent patent box, transfer pricing rules and GST at 18 per cent.

Important Questions

1. Explain the meaning and importance of measuring intangible assets and the legal and practical difficulties involved.

2. Discuss public policy on the measurement of intangible assets and its implications in India.

3. Explain intellectual capital accounting as a management tool with reference to the main frameworks.

4. Explain the methods of extracting profits from intellectual capital and the legal limits on them.

5. How can a start up measure and value its intellectual capital?

6. Explain the IP portfolio as a competitive tool and the role of the intellectual asset manager.

7. Explain IP protection strategy and the choice between patent and trade secret.

8. Explain IP categorisation and the portfolio matrix.

9. Distinguish evaluation and valuation and explain the main valuation approaches.

10. Explain the IP value index, empirical models and option based models.

11. Explain how intellectual capital is measured and monitored.

12. Explain the role of knowledge assets in competitive advantage and product development.

13. Explain patent stocks and copyright collective management in India.

14. Explain the role of intellectual capital in valuing knowledge companies.

15. Compare radical, disruptive, sustaining and incremental innovation.

16. Explain open innovation, its models and its relationship with IP.

17. Explain the importance of patent databases for IP management.

18. Explain IP due diligence and the factors in drafting a licence agreement.

19. Explain NPV, benchmarking and discounted valuation methods with examples.

20. Explain IP as collateral, IP audit and the taxation of IP transactions in India.